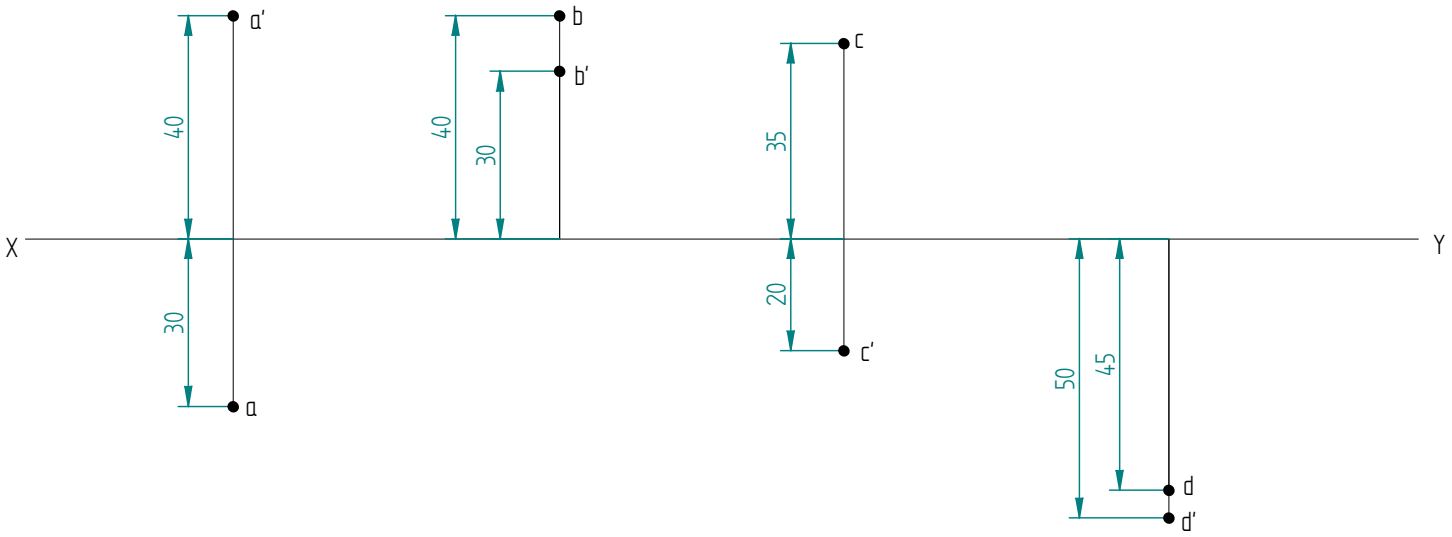
1.1a Draw the projections of the following points on the same XY line.

a) A is 30 mm in front of VP and 40 mm above the HP.

b) B is 40 mm behind the VP and 30 mm above the HP.

c) C is 35 mm behind VP and 20 mm below the HP.

d) D is 45 mm in front of VP and 50 mm below the HP.



A is First Quadrant

B is Second Quadrant

C is Third Quadrant

D is Fourth Quadrant

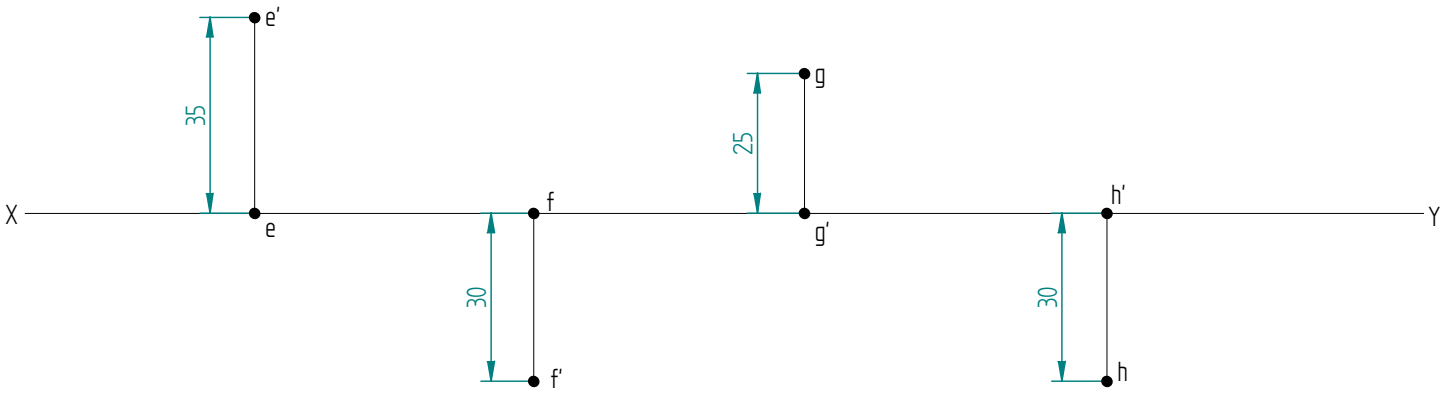
1.2a Draw the projections of the following points on the same XY line.

i) E - 35mm above HP & on VP.

ii) F - 30mm below HP & on VP.

iii) G - On HP & 25mm behind VP.

iv) H - On HP & 30mm in front of VP.



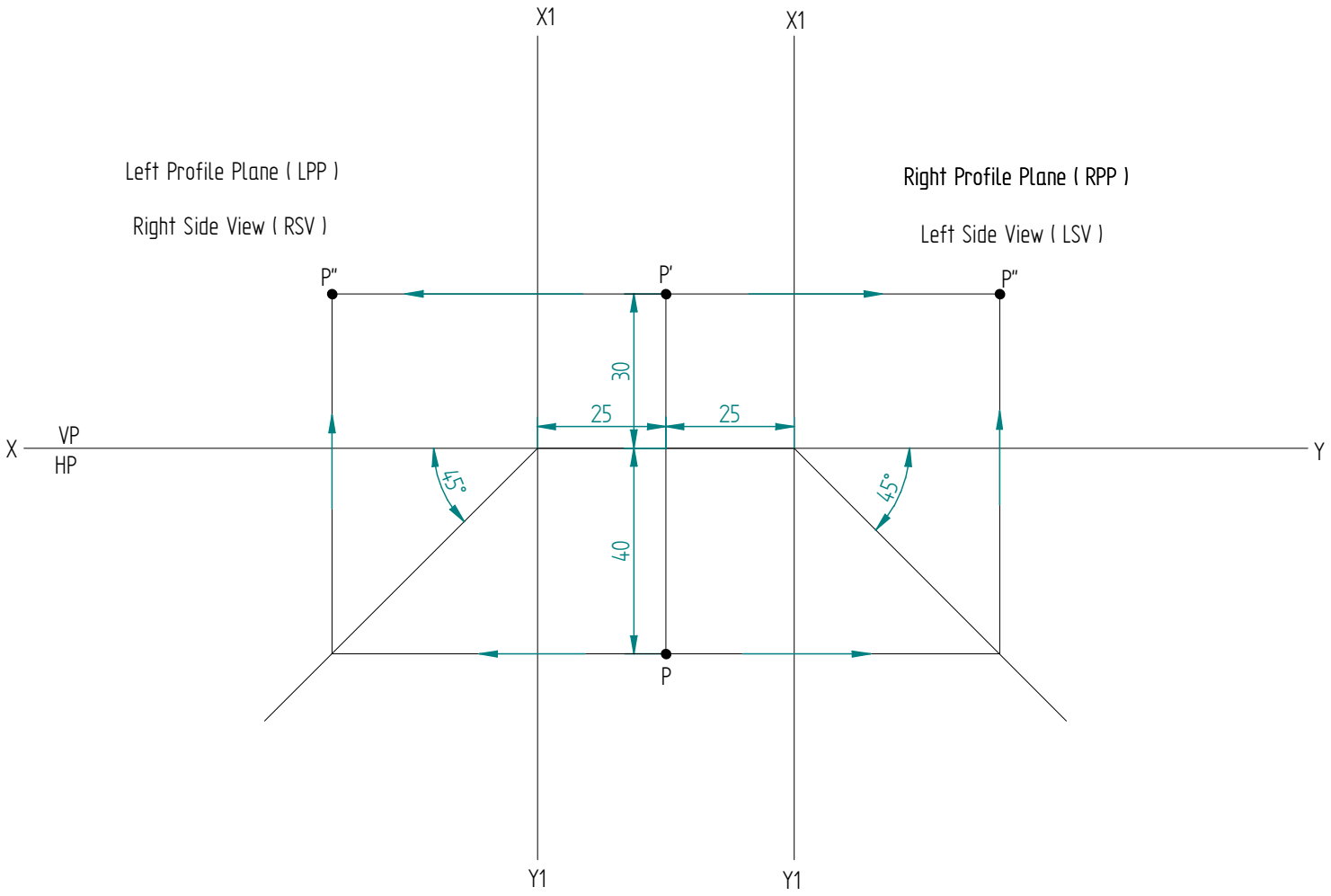
E Is I as well as II Quadrant

F is III as well as IV Quadrant

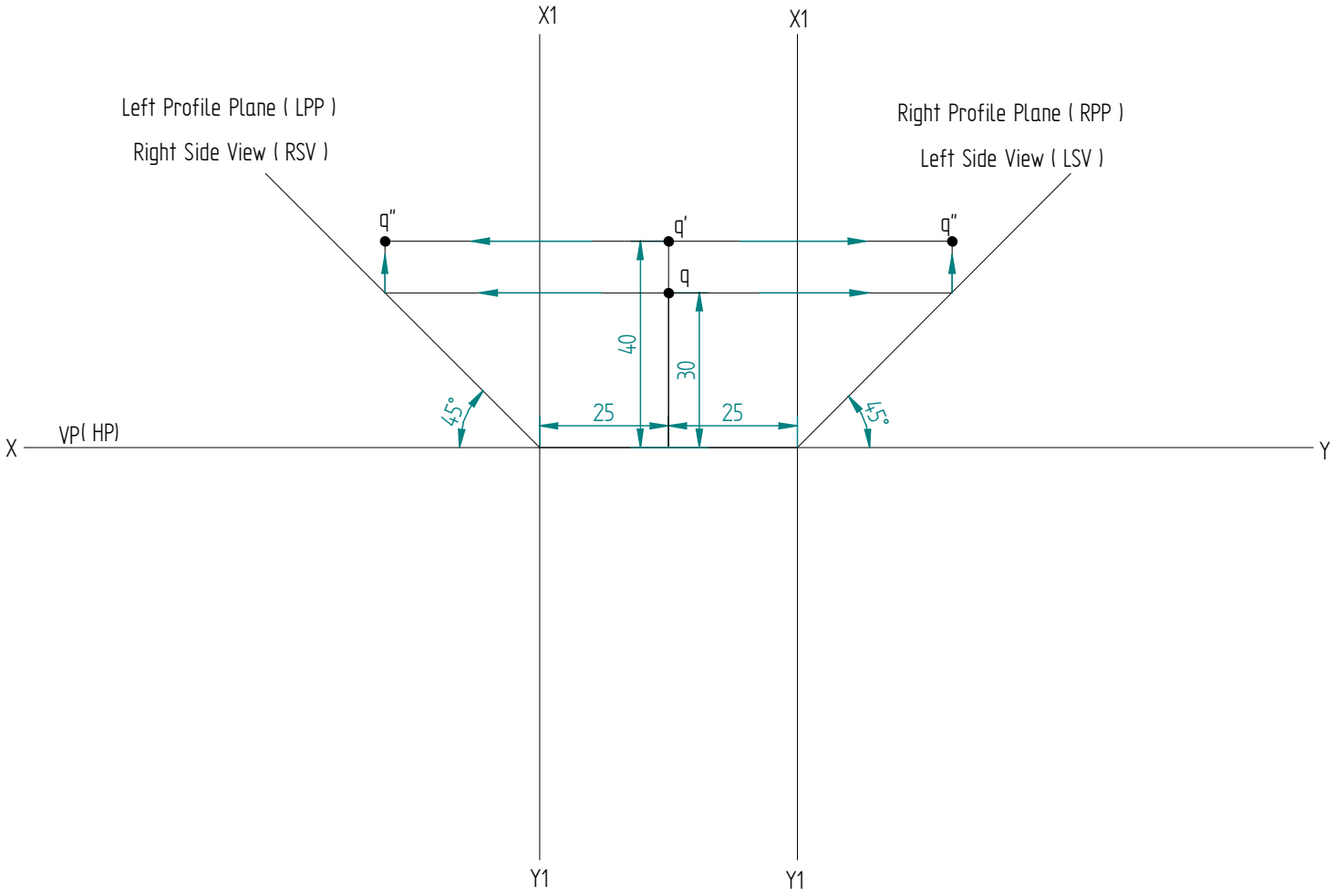
G is II as well as III Quadrant

H is I as well as IV Quadrant

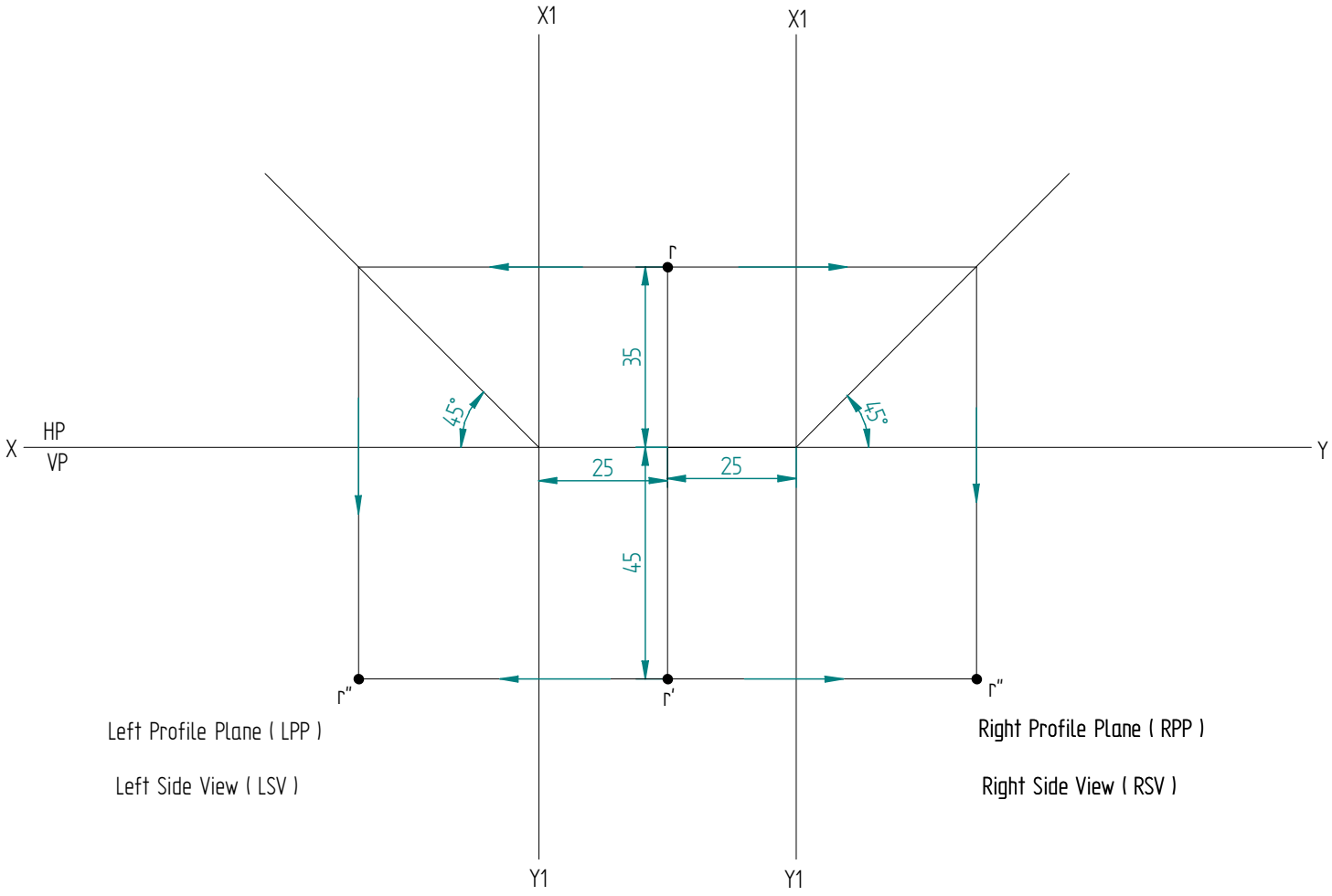
1.3a A Point P is 40 mm in front of VP, 30 mm above HP and 25 mm in front of RPP (Right Profile Plane) / LPP (Left Profile Plane). Draw its projections and name the side view.



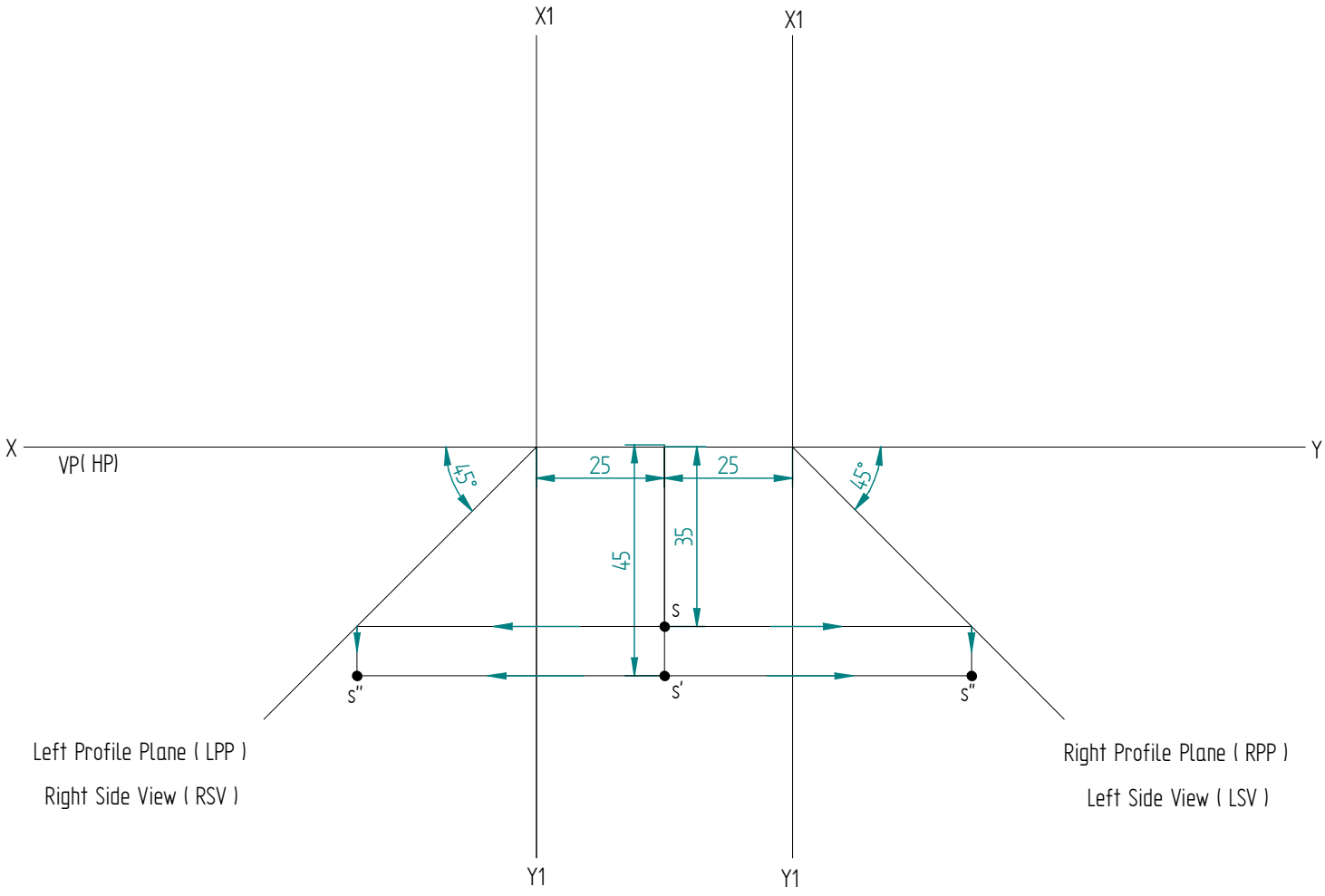
1.4a A Point Q is 30 mm behind VP, 40 mm above HP and 25 mm in front of RPP (Right Profile Plane) / LPP (Left Profile Plane). Draw its projections and name the side view.



1.5a A Point R is 35 mm behind VP, 45 mm below HP and 25 mm behind RPP (Right Profile Plane) / LPP (Left Profile Plane).Draw its projections and name the side view.

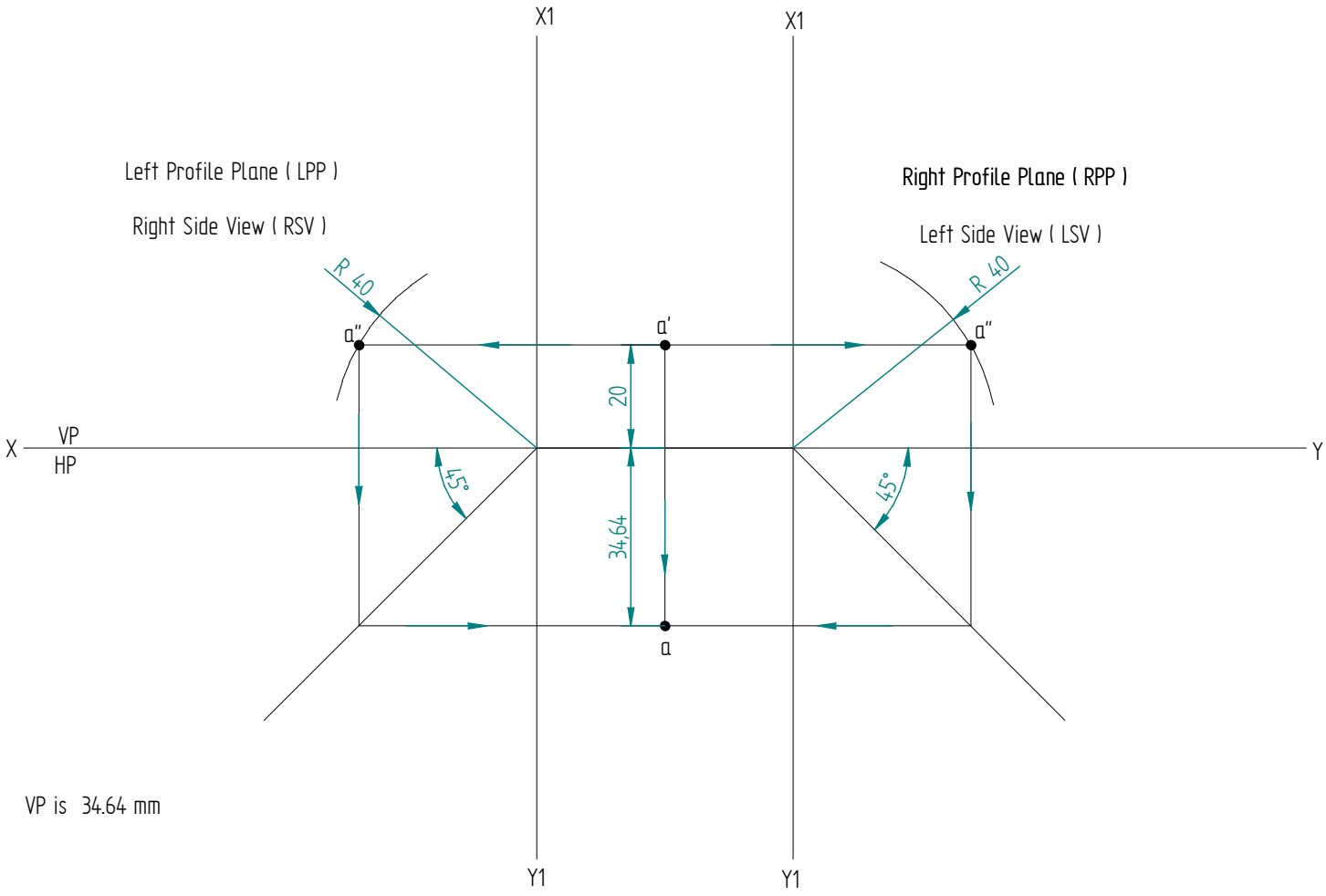


1.6a A Point S is 45 mm in front of VP, 35 mm below HP and 25 mm in front of RPP(Right Profile Plane) / LPP (Left Profile Plane).Draw its projections and name the side view.



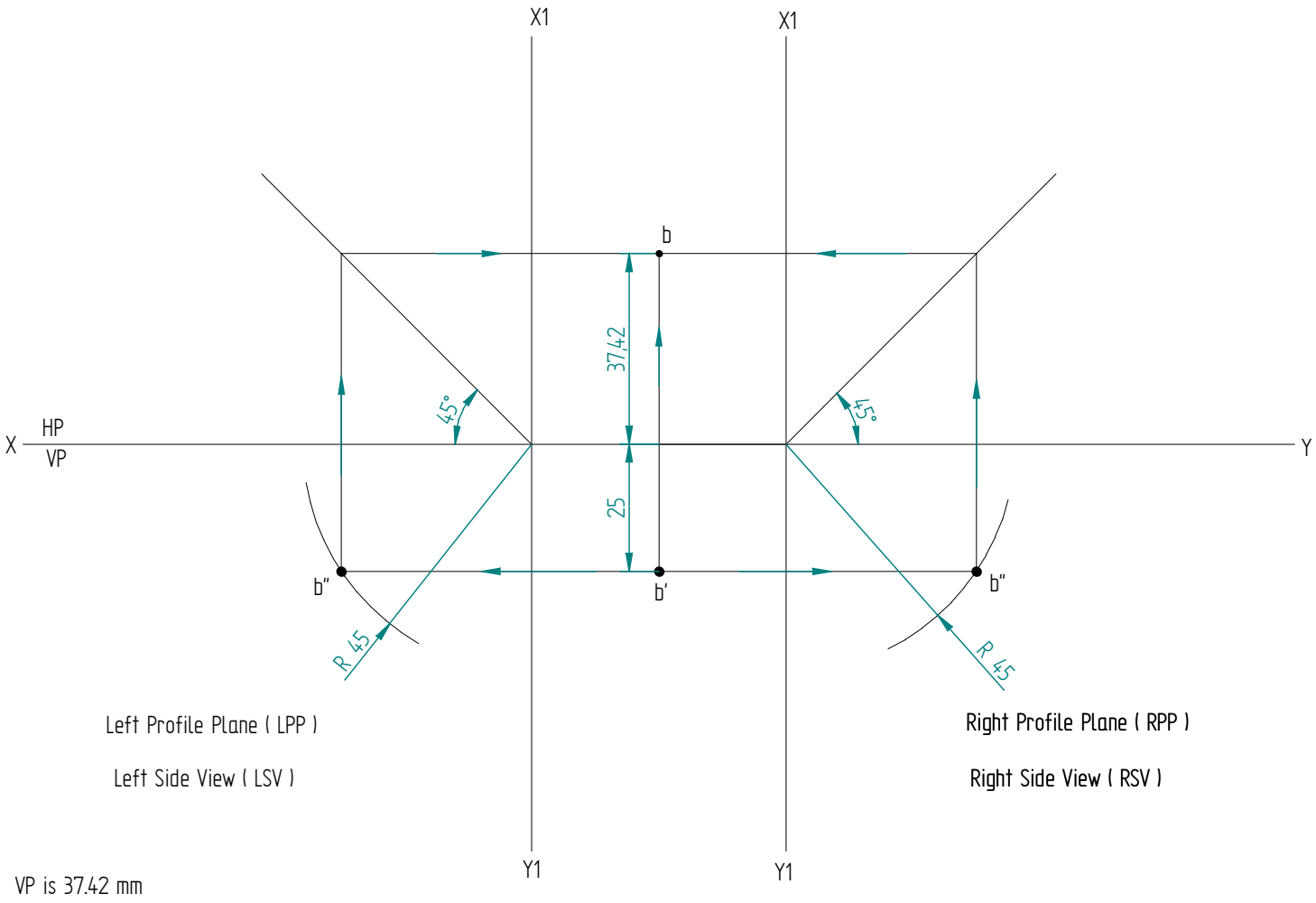
1.7a A point A is 20mm above HP and in the first quadrant its shortest distance from the XY line is 40mm.

Draw the projections. Determine the distance from VP.



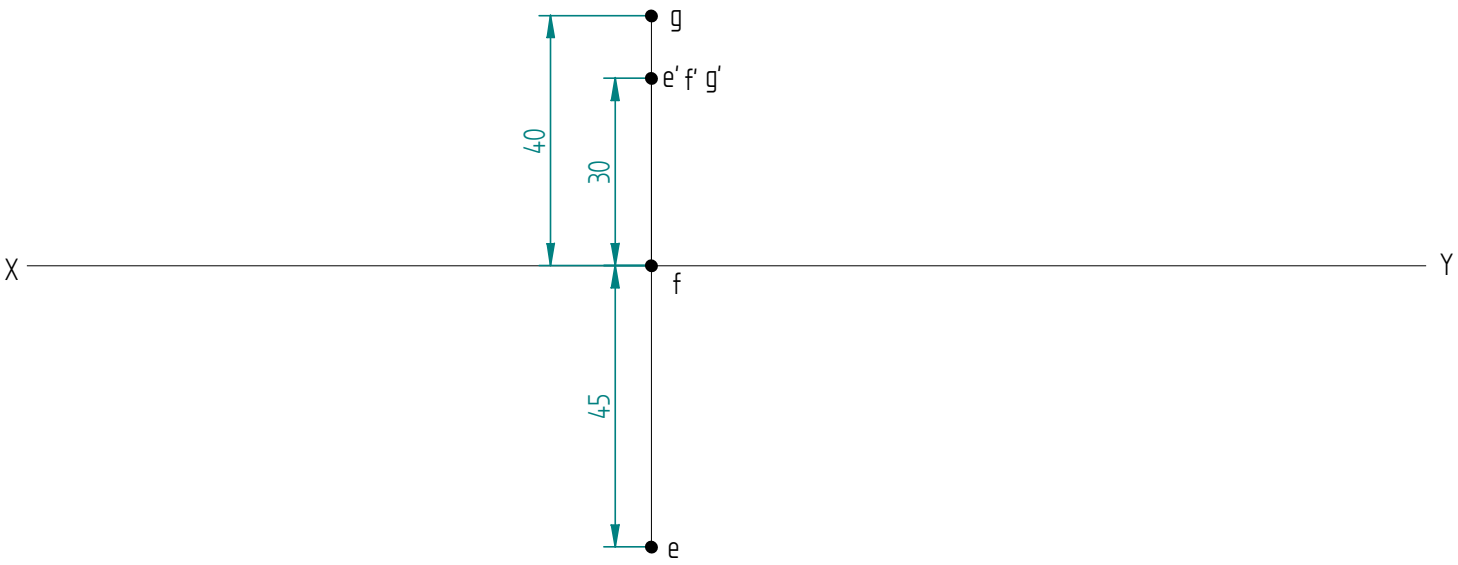
VP is 34.64 mm

1.8a A Point B is 25 mm below HP and is situated in the third quadrant. Its shortest distance from XY line is 45 mm. Draw its projections and find its distance from VP.



VP is 37.42 mm

1.9a A point 30mm above XY line is the front view of three points E, F and G. The top view of G is 40 mm behind VP, the top view of F is on XY line and top view of point E is 45mm in front of VP. Draw the projections of the points & state the Quadrants in which the points are situated.

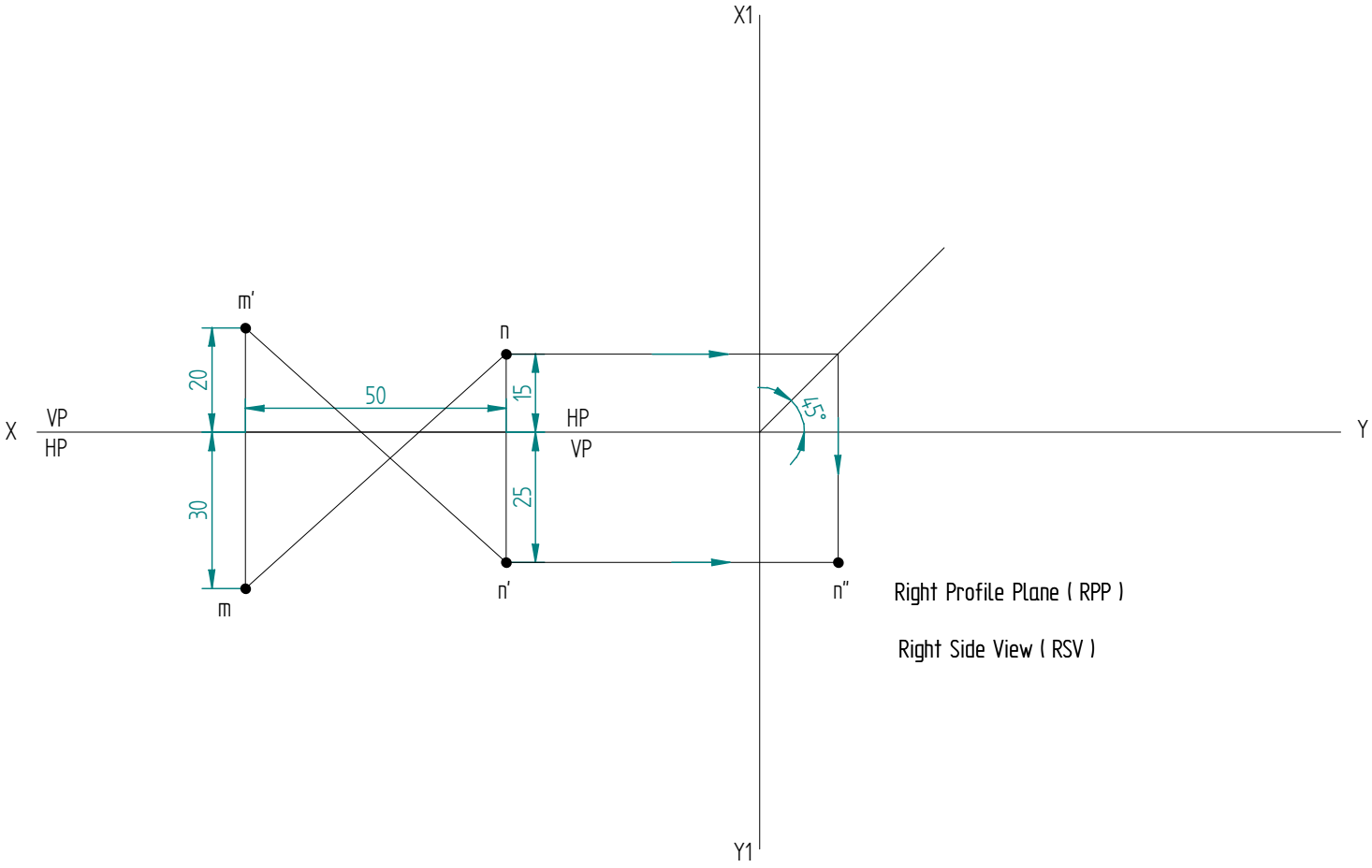


E is I Quadrant

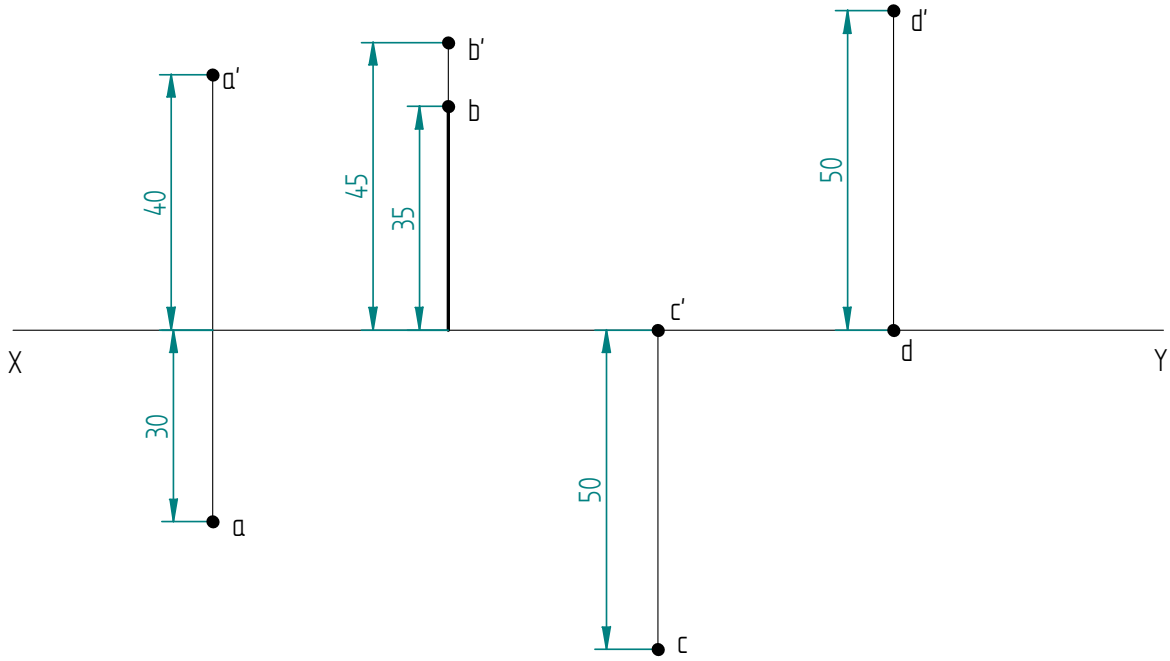
F is I as well as II Quadrant

G is II Quadrant

1.10a A point M is 30mm in front of VP and 20mm above HP. Another point N is 15 mm behind VP & 25mm below HP. The horizontal distance between the points parallel to XY line is 50mm. Draw the projection of the points M & N and join their front and top views. Draw the right side view for the point N only.



1.11a



Answer:

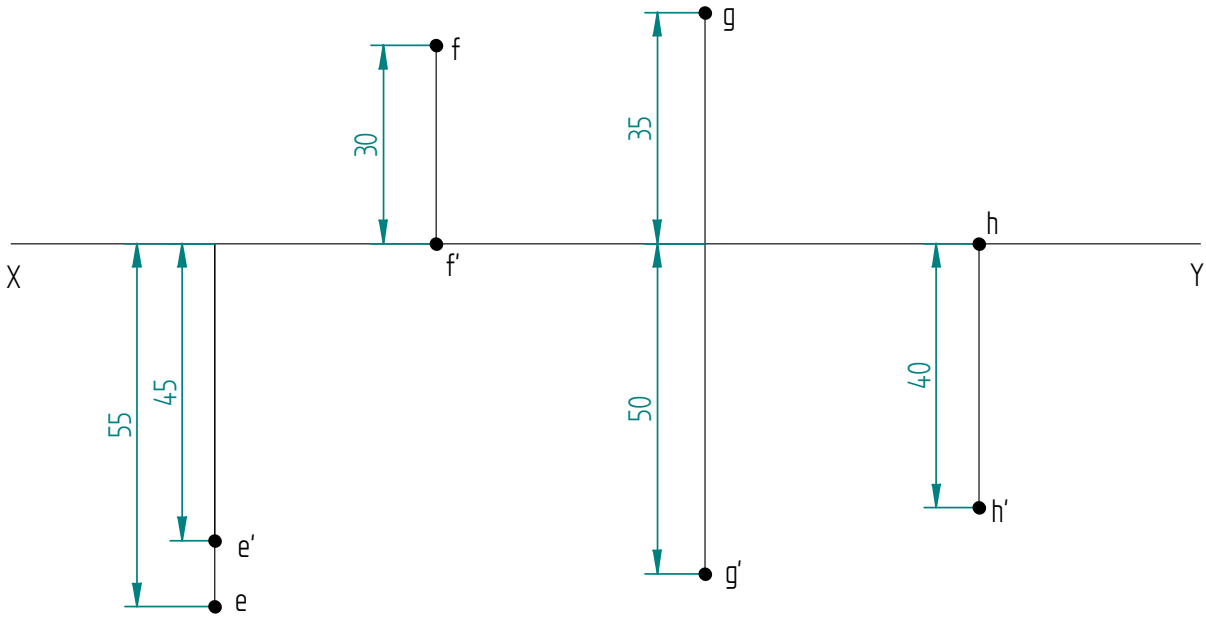
i) A is 40mm above HP & 30mm in front VP (I Quadrant)

ii) B is 45mm above HP & 35mm behind VP (II Quadrant)

iii) C is on HP & 50mm in front VP (I as well as IV Quadrant)

iv) D is 50mm above HP & on VP (I as well as II quadrant)

1.12a



Answer:

i) E is 55mm below HP & 45mm in front VP (IV Quadrant)

ii) F is on HP & 30mm behind VP (II as well as III Quadrant)

iii) G is 50mm below HP & 35mm behind VP (III Quadrant)

iv) H is 40mm below HP & on VP (III as well as IV quadrant)